OMESH SENGAR

Bengaluru, Karnataka | +91 9880295808 | omesh@omeshsengar.com | github.com/ByteWizard-tech

PROFESSIONAL SUMMARY

Computer Science Engineering student (GPA 9.23/10) with hands-on experience across embedded/IoT systems, applied machine learning, and full-stack development. Author on two research manuscripts, including one under revision for Springer SNAM, and an award-winning IoT research paper. Builds complete systems end to end, from sensor hardware and data pipelines to production dashboards, rather than isolated proofs of concept.

EDUCATION

RV College of Engineering (RVCE) *Bengaluru, Karnataka*

B.Tech in Computer Science Engineering — GPA: 9.23/10 *Sep 2024 – Sep 2028 (Expected)*

Relevant Coursework: Data Structures & Algorithms, Discrete Mathematics, Computer Networks, Data Science for Engineers

TECHNICAL SKILLS

Languages: Python, C, JavaScript, TypeScript

Web & Backend: React.js, Next.js, FastAPI, Node.js, HTML5, CSS3, Chart.js, Responsive Design

Machine Learning: Random Forest, Ensemble Learning, Gaussian Process (Normative) Modeling, CNNs, LSTM, Reinforcement Learning (TD3), K-Means Clustering

Embedded / IoT: ESP32-WROOM, FreeRTOS, Sensor Integration (INA219, ADXL345, TMP117), Azure IoT Hub

Data & Algorithms: NetworkX, Graph Theory & Centrality Analysis, Neo4j, Dijkstra's Algorithm, Dynamic Programming (0/1 Knapsack), TSP Heuristics

Cloud & Tools: Supabase, Git, GitHub, MediaPipe

RESEARCH EXPERIENCE

SENTINEL — Autonomous Agentic Framework for Detecting Inciting Social Media Content

Researcher · Manuscript in revision for Springer's Social Network Analysis and Mining (SNAM) journal

- Designed an autonomous agentic detection pipeline combining ADB-based UI automation, LLM-based content scoring, and Neo4j graph-centrality analysis to flag inciting social media content
- Led a major revision addressing 11 reviewer points, resolving an architecture–implementation discrepancy and anonymization/ethics concerns raised in peer review
- Conducted a gap analysis and proposed graph-theoretic feature extensions — multi-centrality, temporal centrality, motif counting, and Louvain community detection

Two-Stage Framework for ASD Detection and Severity Quantification Using Ensemble Learning and Normative Modeling

Researcher, RVCE

- Built a two-stage ML framework achieving 87.83% classification accuracy on resting-state fMRI data for autism spectrum disorder detection
- Extended the pipeline with Gaussian Process-based normative modeling to enable continuous severity estimation beyond binary classification

Hybrid Multi-Scale Framework for Precision Agriculture: Multispectral Remote Sensing with Deep Learning

Researcher, RVCE

- Built an AI-driven crop-diagnosis system fusing multispectral remote sensing with CNN models, achieving 82–96.68% accuracy across multiple crop types
- Deployed real-time disease detection to end users via a mobile application

TECHNICAL PROJECTS

Kinetic Industrial — Predictive Maintenance Sensor Node & Cloud Dashboard

- Engineered an ESP32-based sensor node (ADXL345, INA219, TMP117) with a Random Forest classifier to detect stepper-motor load states, reaching 76.8% cross-session accuracy and 100% stall recall
- Built a companion real-time cloud dashboard using Next.js 16, Supabase, and Azure IoT Hub; delivered a formal engineering report and product poster

Fairpay — Financial Management Application

- Built a JavaScript financial-tracking application that reduced transaction errors by 40% and cut reconciliation time by 50% via a secure, real-time monitoring UI

AWARDS & HONORS

Best Paper Award — ICTST 2025, IFRRA International Conference on Trends in Science and Technology *30 Jul 2025*

- Awarded for the research paper "IoT-Integrated Smart Lab Environment Monitor," presented at an international conference (India–Malaysia collaboration)